FinSuite

A web platform for financial analysis, live market data and next day price forecasting

Technical documentation and evaluation | Version 1.0 | October 2026
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Live system: https://finsuite-py-6ol4.vercel.app/ | **Code:** github.com/purvakakade450/finsuite-py

Abstract

FinSuite is a free website that brings 17 finance tools together: ratio calculators, loan and credit checks, goal planning, live stock prices and news, and a next day price forecast. This document explains how the system is built, where its data comes from, how every result is calculated, and how we checked that the results are correct. It also reports a small forecasting experiment. Two machine learning models were tested on four instruments, and neither beat the simplest possible guess ("tomorrow's price will equal today's") in a consistent way. We regard this as a useful and honest finding, and the website states it openly to its users.

Keywords: personal finance tools, open data, stock market data, price forecasting, Random Forest, system architecture, software verification

1. Introduction

1.1 The problem

Many people decide about loans, savings and shares without the tools that professionals use. Existing tools are often paid, need accounts or API keys, or give an answer without showing how it was reached. A beginner cannot tell whether a number is right, or what it means.

1.2 Aims and design principles

- **Free and keyless.** Anyone can run it. All live data comes from public sources that need no sign up.
- **Explainable.** Every tool states its formula, and a Knowledge Center explains each term in plain words with an example.
- **Correct.** Each calculation is checked against hand computed values, including edge cases.
- **Honest about limits.** Where a result is weak, such as price forecasts, the system says so.
- **Private.** No accounts and no database. Nothing a user types is stored on the server.

1.3 What this document contributes

- A description of the architecture and the live data pipeline, written so that a non specialist can follow it.
- The exact formulas and decision rules behind every score and recommendation.
- A forecasting experiment that compares two models with a baseline, with its limits stated.
- A verification record: how the system was tested and which defects were found and fixed.
- Instructions to run and deploy the system again.

2. System overview

The website has 17 modules, each on its own tab. They fall into four kinds: **Calculation** (formulas on the numbers a user types), **Rule based** (fixed scoring rules), **Live data** (Yahoo Finance and Google News), and **ML** (machine learning).

#	Module	What it does	Kind
01	Ratios	ROE, ROA and debt ratio, with a health label.	Calculation
02	Finance Report	Income, expenses, savings and debt turned into ratios and tips.	Calculation
03	AI Assistant	Explains finance terms from built in definitions and gives a quick snapshot. It does not call an outside AI service.	Rule based
04	Loan Insights	Comments on each customer's credit profile in a loan table.	Rule based
05	Credit and Risk	Credit score category and risk level, adjusted for the EMI burden.	Rule based
06	Investment Call	BUY, HOLD or AVOID from four checks.	Rule based
07	Company Performance	Score out of 100 from growth, customer satisfaction and staff retention.	Rule based
08	Portfolio Dashboard	Scores and ranks several companies.	Rule based
09	Goals and Retirement	Monthly saving needed for a goal, allowing for inflation; emergency fund check.	Calculation
10	Knowledge Center	67 plain language explanations with examples, grouped in six sections, with search.	Reference
11	Stock Market	Search any listed company; live quote, daily history table and chart, news.	Live data
12	Growth and Target	What an earlier investment is worth now; target and stop loss planning.	Calculation
13	Portfolio and Tax	SIP versus lump sum, capital gains tax estimate, 52 week range position.	Calculation
14	Watchlist and Alerts	Live prices for chosen stocks and price alerts.	Live data
15	Data Sources	Says where the information comes from.	Reference
16	About	Project, designer and guide.	Reference
17	ML Prediction	Next day close forecast from two models, shown beside a baseline.	ML

3. Architecture

FinSuite has four layers. The **browser** shows pages and sends requests. The **application server** receives them and checks the inputs. The **services** do the real work: formulas, market data and forecasting. The services fetch outside data only from two public sources.

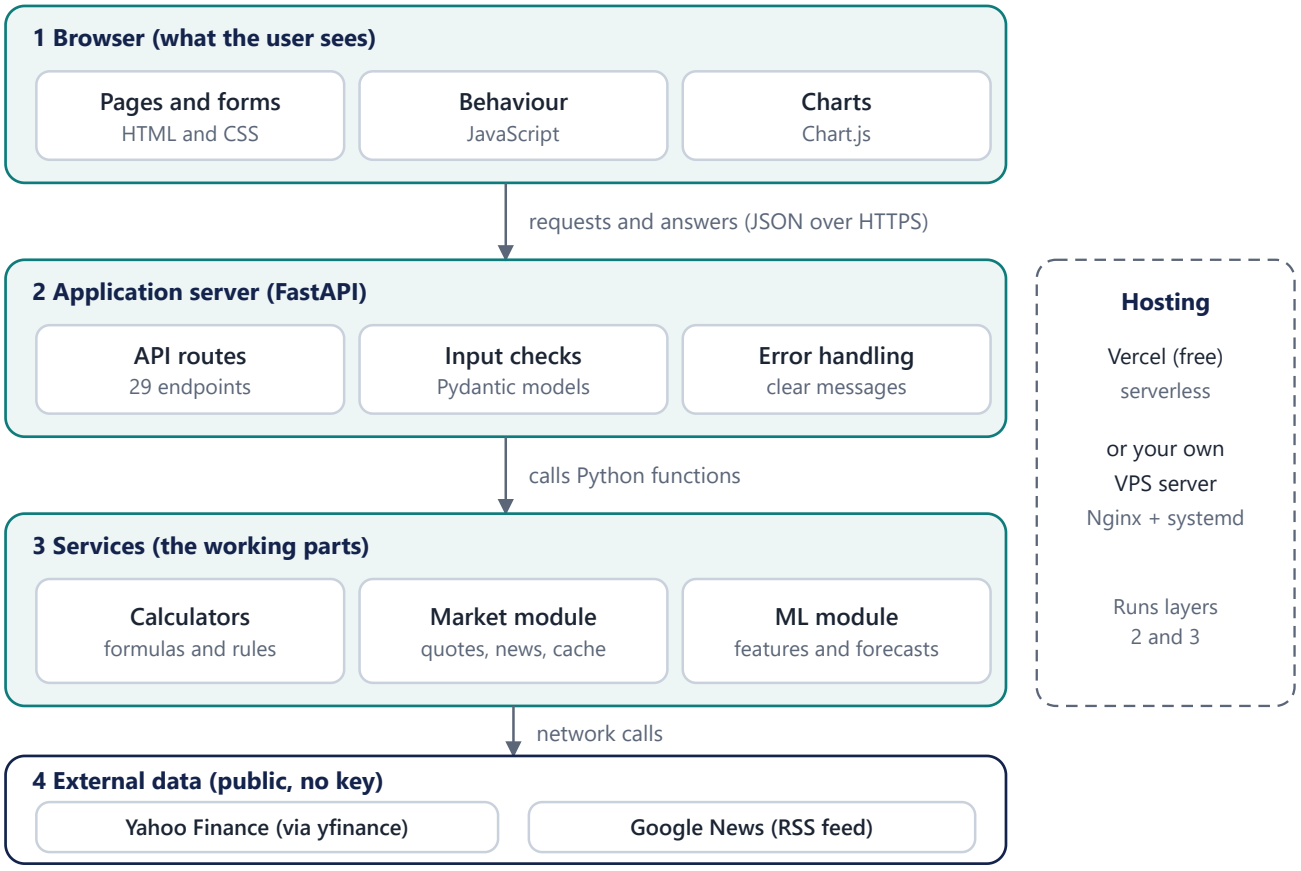


Figure 1. System architecture. Requests go down the layers; answers come back up the same way.

3.1 The layers in plain words

- **Browser.** Plain HTML, CSS and JavaScript, with the Chart.js library for charts. No framework is used, which keeps the code small and readable.
- **Application server.** Written in Python with FastAPI. It exposes 29 endpoints (web addresses that each do one job). It checks every input and turns problems into clear messages instead of crashes.
- **Services.** Three Python modules: the calculators, the market module (search, quotes, history, news, short term memory) and the ML module (features and models).
- **External data.** Yahoo Finance, read through the `yfinance` library, and the public Google News RSS feed.

3.2 Design decisions

Decision	Reason	Trade off
Python and FastAPI	Finance maths and machine learning use the same language; inputs are validated automatically.	Needs a Python runtime to host.
No framework in the browser	Small, no build step, easy for a beginner to read.	More manual code for the page.
No database or login	Nothing personal is stored; simple and private.	The watchlist and alerts are lost when the page reloads.
Short term memory (cache)	Protects the free data sources and makes repeat requests fast.	Memory is cleared when the server restarts.

Public data, no keys	Anyone can reproduce the system at no cost.	Sources are unofficial: data can be delayed or limited.
Version stamp on page files	The browser never mixes a new page with an old script after an update.	None of note.
Ignore out of date replies	If a user picks a second stock quickly, a slow reply for the first is discarded.	Small extra code in the page.

4. How data is collected

4.1 Sources

Source	Data	How it is read	Used by
Yahoo Finance	Company search, latest price, daily open, high, low, close and volume	`yfinance` library, no key	Modules 11, 14, 17
Google News	Headlines from the last 7 days	Public RSS feed, no key	Module 11
The user	Numbers typed into forms	Sent to the server, not stored	All other modules

4.2 What happens when a quote is requested

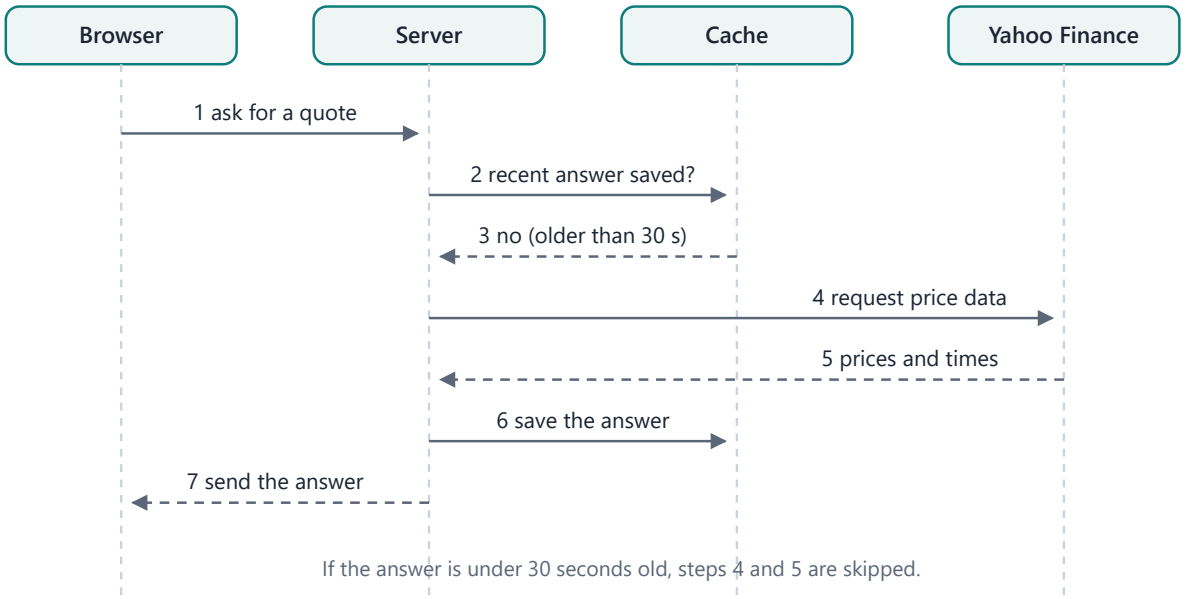


Figure 2. Sequence for one live quote. The cache answers directly when the stored value is less than 30 seconds old.

The price shown is the last traded price. To get it, the server reads daily bars to learn the previous close, then reads the latest one minute bars to find the last trade and its time. On weekends and holidays the most recent session is shown, with its time, so users can tell the market is closed.

4.3 Finding the right symbol

Users may type a company name ("Tata Steel") or a code ("TATASTEEL.NS"). Names are searched first. Codes are tried as typed, then with `.NS` (NSE) and `.BO` (BSE) added. The first guess that returns real price data is

used. If nothing works, a clear "not found" message is shown.

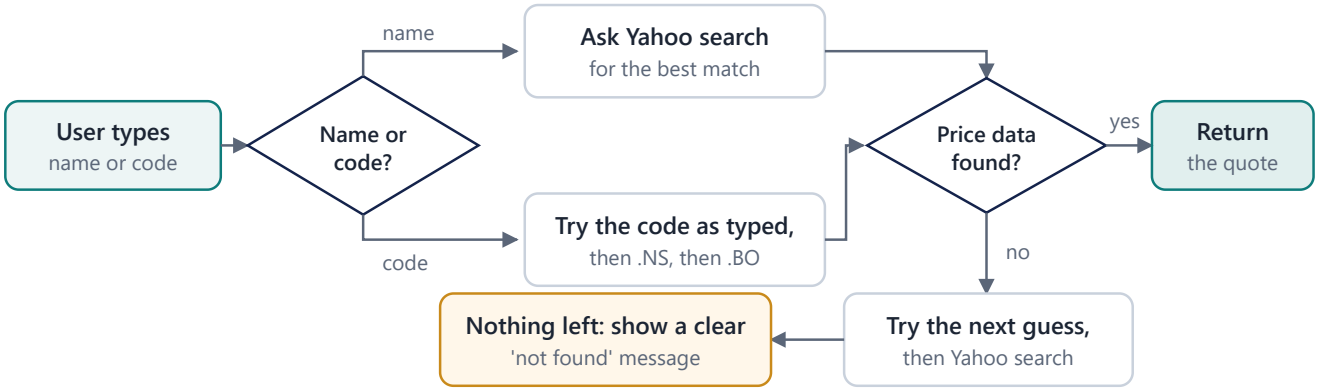


Figure 3. How user input is turned into a symbol that has data.

4.4 Short term memory and fault tolerance

Data	Kept for	Why
Live price	30 seconds	Prices change often, but repeated clicks need no new request.
Daily history	10 minutes	Daily bars rarely change during the day.
News	5 minutes	Headlines change slowly.
Company search	1 hour	Company names do not change.

If a source fails, the last good answer is returned instead of an error. Automatic refresh runs only while the tab is open and visible: quotes every 30 seconds and news every 5 minutes.

4.5 News relevance

Searching news by a company name returns many pages that merely mention it. To avoid this, a headline is kept only if its title contains the company name or its ticker. Repeated copies of the same story are removed and the newest are shown first.

5. Methods

5.1 Calculation tools

Quantity	Formula	Rule or label
ROE	`Net income / Equity x 100`	Above 20% Excellent, 10 to 20% Good, below 10% Weak
ROA	`Net income / Assets x 100`	10% or more treated as strong
Debt ratio	`Liabilities / Assets`	Higher means more borrowing
Debt to equity	`Debt / Equity`	Above 1: owes more than owns
Current ratio (personal)	`Savings / Monthly expenses`	Months of expenses covered

Savings ratio	`Savings / Income x 100`	Below 20% flagged as thin
Risk reward	`(Target - Entry) / (Entry - Stop)`	2 or more favourable; 1 to 2 modest; below 1 weak
Annual growth (CAGR)	`(Value now / Amount)^(1/years) - 1`	Shown for holds of about a month or more
52 week position	`(Price - Low) / (High - Low)`	0% at the low, 100% at the high
Emergency fund	`Savings / Monthly expenses`	6 months or more protected; 3 to 6 building; under 3 critical

5.2 Goal planning

A target given in today's money is first raised by inflation: `Future target = Target x (1 + inflation)^years`. Existing savings are grown at the expected monthly return, and the monthly investment is the amount that closes the remaining gap. With monthly rate `i = return / 12` and `n = months`, the monthly amount is:

```
SIP = Remaining target / [ ((1 + i)^n - 1) / i x (1 + i) ]
```

Both the savings and the SIP use the same monthly compounding, so the yearly schedule ends exactly on the target. (Our tests found that mixing yearly and monthly compounding broke this; see Section 6.)

5.3 Capital gains tax (India, listed shares)

Holding for 365 days or less is short term and taxed at 20% of the gain. Holding for longer is long term, taxed at 12.5% on the part of the year's gains above Rs 1,25,000. A 4% cess is added to the tax. A loss is not taxed. These rules were applied as modelled here and may change; the tool is an estimate, not tax advice.

5.4 Rule based scores

These scores are expert rules with conventional thresholds. They were not fitted to data. They are meant to teach, and each rule is shown to the user.

Score	Rule	Result
Credit category	Score bands: below 550, 550, 650, 750, 800	Poor, Fair, Good, Very Good, Excellent
Credit risk level	Base level from the band; moves up one step if EMI is above 40% of income	Very High to Very Low risk
Investment call	One point each: revenue growth 15% or more, profit margin 15% or more, ROA 10% or more, ROE 15% or more	3 or 4 BUY; 2 HOLD; below 2 AVOID
Portfolio score	ROE above 15% (30 points), ROA above 10% (25), growth above 10% (25), margin above 15% (20); scaled to the figures given	75+ BUY, 50 to 74 HOLD, below 50 SELL
Company performance	Growth above 15% (34), satisfaction above 80% (33), retention above 85% (33)	85+ Excellent, 60+ Good, else Needs Improvement

5.5 Machine learning forecast

The ML module predicts the **next trading day's closing price** of a chosen symbol, using only that symbol's own price history.

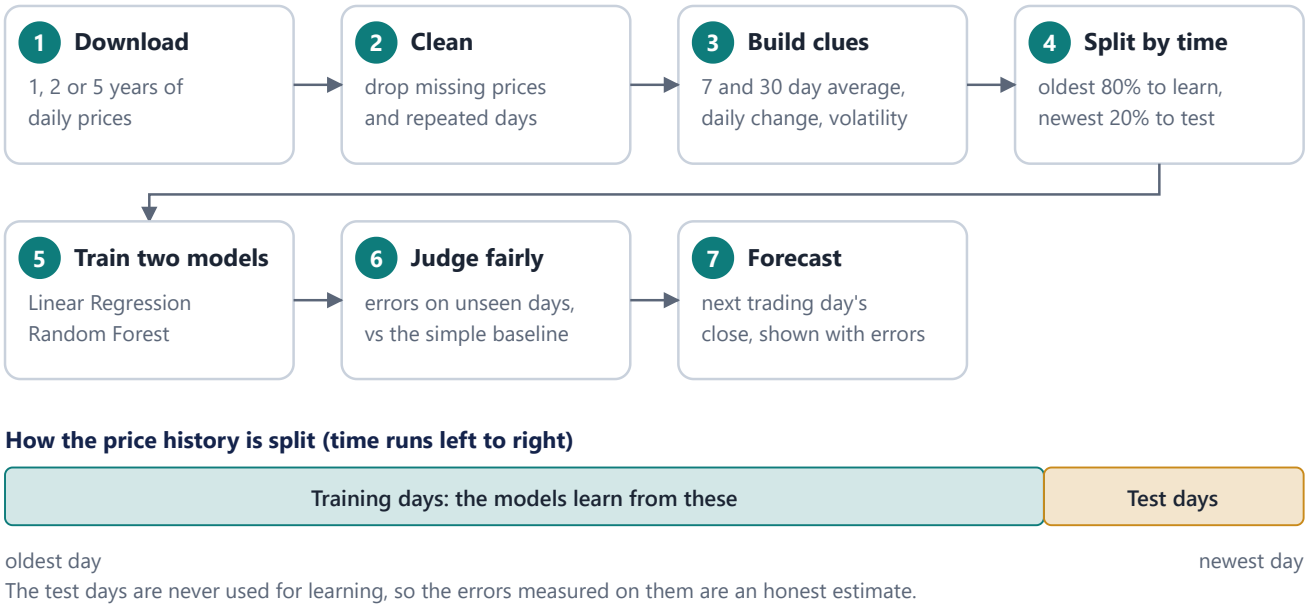


Figure 4. The forecasting pipeline and the time ordered split.

- **Data.** 1, 2 or 5 years of daily bars. At least 120 trading days are required.
- **Features (the clues).** Today's close; the average close of the last 7 and 30 days; today's percentage change; and volatility, meaning the spread of daily changes over the last 21 days.
- **Target.** The next day's close.
- **Split.** By time, never shuffled: the oldest 80% of days are for learning and the newest 20% are for testing. Shuffling would let a model "see the future" and give false accuracy.
- **Models.** Linear Regression on standardised features, and a Random Forest (200 trees, depth 8). The forest predicts the *change from today's close* and adds it back.
- **Baseline.** The naive guess "tomorrow equals today". A model is useful only if it beats this.
- **Errors.** MAE (average miss in price), RMSE (like MAE but punishes big misses) and MAPE (average miss as a percentage). Lower is better.

> **A design correction.** A first Random Forest that predicted the price level directly did poorly: on TCS its MAE was 112.16 against 35.21 for the baseline. Trees cannot predict a price outside the range seen in training, and prices drift upward. Predicting the day's change instead reduced the MAE to 47.30. The cause is the model's structure, not the data.

6. Verification

6.1 Approach

- **Hand computed checks.** For each calculator, the expected answer was worked out independently and compared, including boundaries (for example a credit score of 549 versus 550, or 749 versus 750).

- **Invalid input checks.** Zero divisors, out of range values and impossible dates must give a clear message, never a crash.
- **Consistency checks on live data.** Price change must equal price minus previous close; the price must lie within the day's range; each day's high must not be below its open or close.
- **Goal schedule check.** The last point of the yearly schedule must equal the inflation adjusted target.
- **End to end browser tests.** All 17 tabs were exercised in a real browser against the live site, including error messages and a phone sized screen.

More than 100 automated checks were run and repeated after every change. For one instrument (Apple), the reported errors were also recomputed independently from the returned predictions and matched.

6.2 Defects found and fixed

Area	Defect found	Fix
Goal planner	With existing savings, savings grew yearly but the SIP monthly. At year 10 the plan showed Rs 18,29,755 against a Rs 17,90,848 target.	Same monthly compounding for both; schedule now lands on the target.
Goal planner	A 2.5 year plan stopped at year 2 (Rs 8,97,038 against a Rs 11,56,817 target).	Schedule now ends at the true end of the plan.
52 week range	A price below the low showed 2% instead of 0%.	Position limited to 0 to 100%.
Assistant lookup	"Debt to equity" returned the definition of "equity"; "premium" matched "EMI".	Whole word matching, longest phrase first.
SIP and range tools	A zero price caused a server error.	Clear validation message.
Input validation	Negative current price gave a growth of -105%; stop loss above 100% gave a negative stop price; a risk score of 1000 was accepted.	Such inputs are rejected with a message.
ML model	Random Forest could not extrapolate (see Section 5.5).	Predict the change from today's close.
Minor	A trailing comma in the SIP price list; some buttons failed silently; a favicon request returned 404; "15.0 year(s)" in a message.	Each corrected.

7. Results of the forecasting experiment

Setup. Two years of daily history; oldest 80% for training, newest 20% (about 93 trading days) for testing; last trading day 16 October 2026. Four instruments were used: TCS, Reliance (Indian equities), Apple (US equity) and the NIFTY 50 index.

Instrument	Baseline MAE	Linear Regression MAE	Random Forest MAE	Baseline MAPE
TCS (Rs)	35.21	35.67	47.30	1.58%
Reliance (Rs)	13.02	13.08	13.25	1.02%
Apple (USD)	3.97	3.99	4.03	1.27%
NIFTY 50 (points)	134.97	132.04	140.52	0.57%

Table. Mean absolute error (MAE) on the test days. Lower is better.

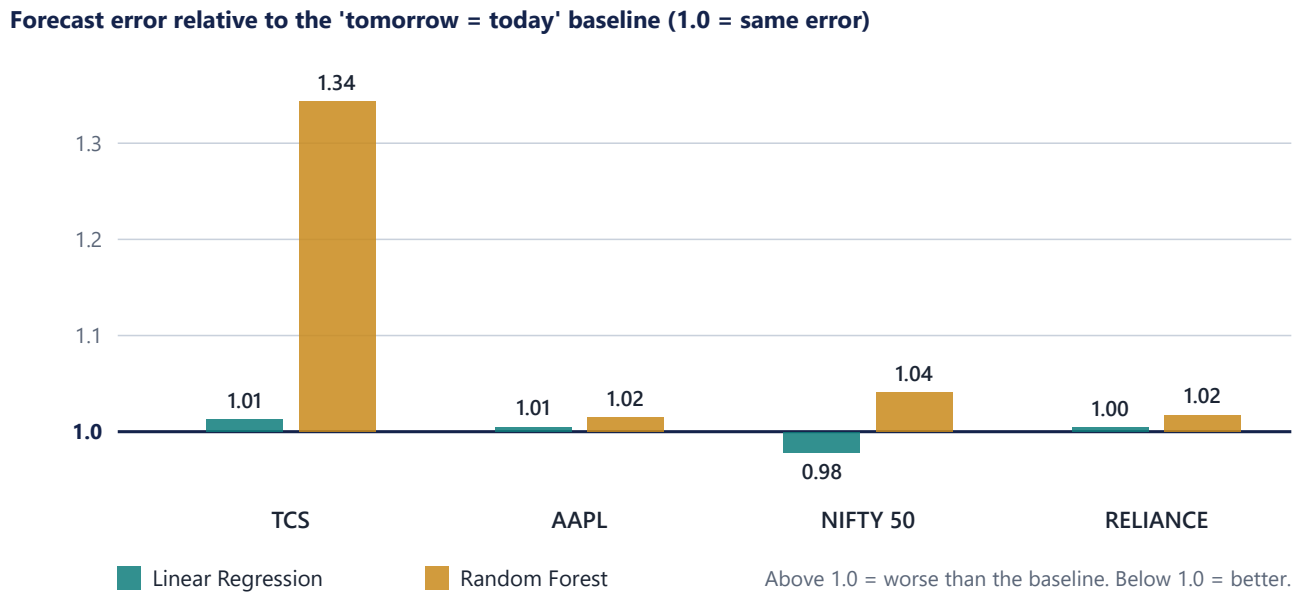


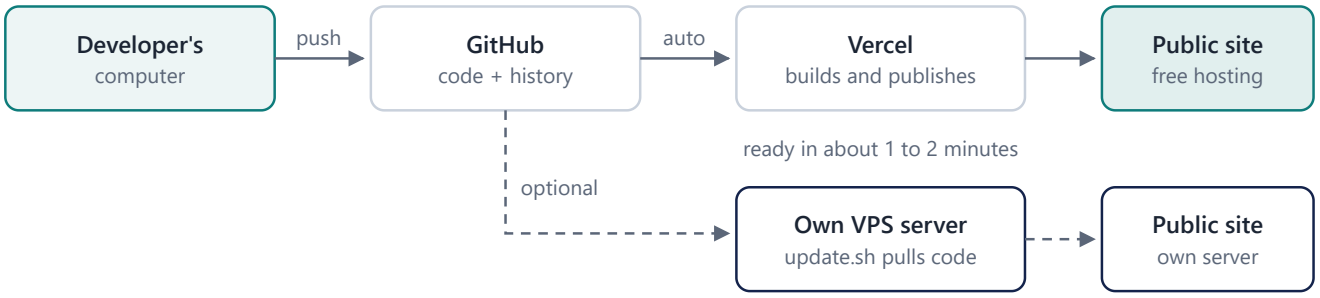
Figure 5. Each model's error divided by the baseline's error. Bars close to 1.0 mean no real difference from the baseline.

7.1 What the results show

- In seven of the eight model and instrument pairs, the model was slightly *worse* than "tomorrow equals today". The one exception was Linear Regression on the NIFTY 50, which was about 2% better.
- Six of the eight differences are within 2%, which is small for a single test window. The larger gaps were the Random Forest on TCS (34% worse) and on the NIFTY 50 (4% worse).
- This agrees with the common view that day to day price changes are very hard to predict from price history alone.

> **Limits of this experiment.** One test window of about 93 days per instrument; one run; no statistical test or confidence intervals; four instruments only; no trading costs; models see price history only (no news, results or market wide data). The results show that these two simple models add little, not that no model could.

8. Deployment and how to run



Solid path: used today. Dashed path: ready-made files exist (deploy folder), not yet used.

Figure 6. From the developer's computer to the public site.

The public site runs on Vercel's free plan. Each time the code on GitHub changes, Vercel rebuilds and publishes it automatically in one to two minutes. The free plan is intended for personal, non commercial use, which fits a research project. The ML tab runs on this plan (a forecast took about four seconds in our check). For a dedicated server, the repository includes ready made files for an Ubuntu VPS (a service definition, an Nginx configuration and setup and update scripts) in the `deploy` folder. These were not used for the public site.

8.1 Run it on your own computer

Requirements: Python 3.11 or newer (3.13 was used) and an internet connection for live data.

```
python -m venv agentic_env
agentic_env\Scripts\activate          (Mac or Linux: source agentic_env/bin/activate)
pip install -r requirements.txt
python app/main.py
```

Then open `http://127.0.0.1:8000`. Set the `PORT` variable to use another port. Press Ctrl and C to stop.

8.2 Project layout

Folder or file	Contents
frontend/	The page (index.html), behaviour (app.js) and design (style.css).
app/main.py	The server: all endpoints, error handling, serving the page.
app/calculators.py	All formulas and rules for modules 01 to 09, 12 and 13.
app/market.py	Search, symbol resolution, quotes, history, news and the cache.
app/predict.py and backend/services/	Forecast pipeline: cleaning, features and the two models.
app/glossary.py	The 67 Knowledge Center entries.
deploy/	Optional VPS setup files and guide.
docs/	Project report and this document.

9. Limitations

- **Data quality.** Yahoo Finance data is unofficial, may be delayed, and may be rate limited from some servers. We could not guarantee uptime of free public sources.
- **Rules, not research findings.** Thresholds such as 15% ROE or 40% EMI to income are common rules of thumb. They were not calibrated on data.
- **Tax model.** The capital gains tool covers Indian listed shares only and simplifies the rules.
- **No persistence.** The watchlist and alerts live in the browser tab. Alerts work only while the page is open.
- **Assistant.** It matches keywords to stored definitions; it does not understand free text.
- **Forecasting.** Narrow experiment (Section 7). The forecast should not be used for trading.
- **No user study.** We checked correctness, not how easy the tools are for real users.

10. Future work

Direction	Ideas
Make it a product	Saved watchlist and alerts; user login; real background alerts by email; PDF or Excel download of results.
More tools	Portfolio tracker with live profit and loss; income tax comparison (old and new regime); EMI schedule with prepayment; mutual fund and SIP tracker using public NAV data; budget planner; retirement (FIRE) calculator.
Better analysis	Technical indicators on charts (RSI, moving averages, Bollinger bands); news sentiment linked to prices; stock comparison and currency conversion.
Stronger evaluation	Walk forward testing over many windows, more instruments, significance tests, extra features such as market wide indicators, and a usability study.

11. Glossary

Endpoint	A web address on the server that does one job, for example calculating ROE.
REST / JSON	A common way for programs to exchange requests and answers as simple text.
Cache	Short term memory that avoids asking a source the same thing twice.
RSS feed	A public list of the latest headlines that programs can read.
yfinance	A free library that reads public Yahoo Finance data.
Baseline	The simplest guess a model must beat to be useful.
Serverless	Hosting where the provider starts the program only when a request arrives.
VPS	A rented virtual server that you manage yourself.

12. Links and contact

Live system: https://finsuite-py-6ol4.vercel.app/
Source code: https://github.com/purvakakade450/finsuite-py
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LinkedIn: linkedin.com/in/purva-kakade-356a59237
Guidance: Prof. Ashish Singh, Shailesh J. Mehta School of Management, IIT Bombay (https://www.som.iitb.ac.in/prof-ashish-singh/)

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